REPORTING

Original Reported Date:	JUN-20-2023
Amended Date:	JUN-27-2023
Receipt Date:	JUN-16-2023
Collection Date:	JUN-15-2023
Specimen:	Blood
Status:	AMENDED

PHYSICIAN

Chih-Hsueh Chen
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Additional Recipient: N/A

Complete Tumor Response Map on page 2

Summary of Detected Somatic Alterations, Immunotherapy Biomarkers & Associated Treatment Options

KEY ✔ Approved in indication ~ Approved in other indication ✕ Lack of response

Detected Alteration(s) / Biomarker(s)	Associated FDA-approved therapies	Clinical trial availability (see page 3)	% cfDNA or Amplification
KRAS G12D	None	Yes	16.0%
TP53 Y220C	None	Yes	3.2%
TP53 Y234C	None	Yes	1.2%
TP53 R267W	None	Yes	0.4%
TP53 R249fs	None	Yes	0.3%
TP53 L265del	None	Yes	0.2%

Comments

Amended Report: On the report released JUN-20-2023, the diagnosis was indicated as Lung adenocarcinoma. In this amended report, the diagnosis has been updated to Endometrial carcinoma and the available clinical trials have also been updated. There are no other changes to the report.

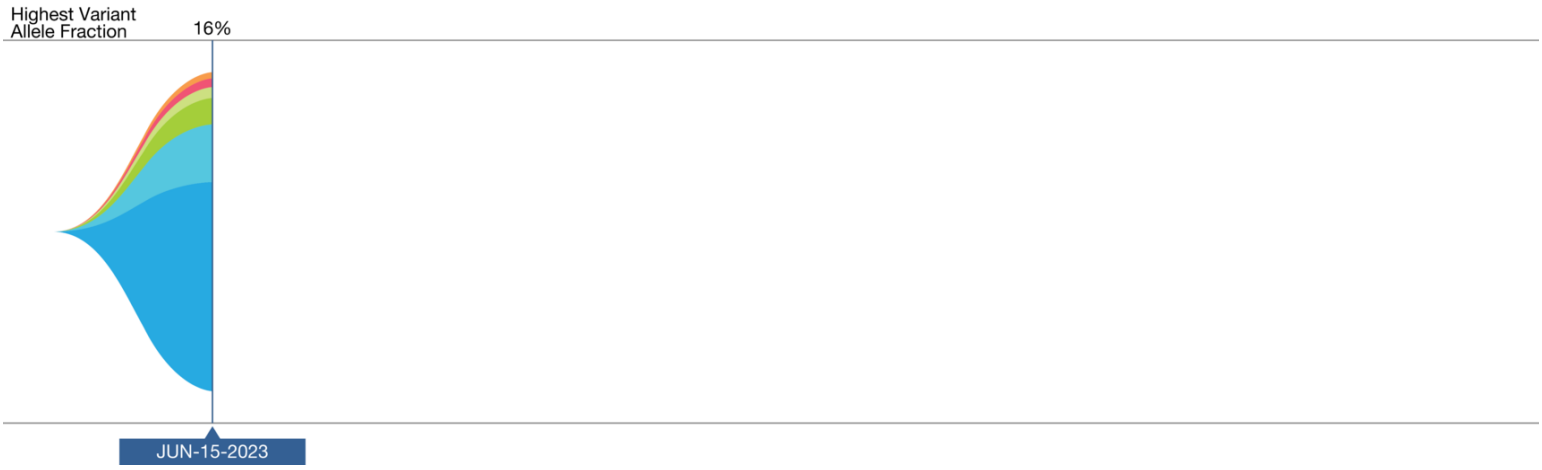
Additional Biomarkers

Biomarker	Additional Details
MSI-High	NOT DETECTED

Alterations or biomarkers that were "NOT DETECTED" have been excluded from the summary table above.

Guardant360 Tumor Response Map

The Guardant360 Tumor Response Map illustrates the variant allele fraction (% cfDNA) of observed somatic variants at each sample submission. Amplifications are not plotted, and only the first and last five test dates are plotted. Please see the Physician Portal (portal.guardanthealth.com) for the Tumor Response Map with all test dates.



Detected Alteration(s) / Biomarker(s)	% cfDNA or Amp
KRAS G12D	16.0%
TP53 Y220C	3.2%
TP53 Y234C	1.2%
TP53 R267W	0.4%
TP53 R249fs	0.3%
TP53 L265del	0.2%

The table above annotates the variant allele fraction (% cfDNA) detected in this sample, listed in descending order.
§ See definitions section for more detail

Available Clinical Trials (within the same state as the ordering physician)

There may be additional trials not listed here. Visit: **portal.guardanthealth.com** or email **clientservices@guardanthealth.com** with A0779678 in the subject line of the email, for additional trials.

Alteration	Trial ID / Contact	Title	Phase	Site(s)
TP53 Y220C	NCT04768868 Jian Wang,Jian. wang@impacttherapeutics.com,+86 18613056501	The Safety and Pharmacokinetics Preliminary Efficacy of IMP7068 in Patients With Advanced Solid Tumors	Phase 1	Taipei, Taiwan Taoyuan, Taiwan Taichung, Taiwan Tainan, Taiwan (2)
	Visit **portal.guardanthealth.com** for trials not within the same state as the physician's office			
TP53 Y234C	NCT04768868 Jian Wang,Jian. wang@impacttherapeutics.com,+86 18613056501	The Safety and Pharmacokinetics Preliminary Efficacy of IMP7068 in Patients With Advanced Solid Tumors	Phase 1	Taipei, Taiwan Taoyuan, Taiwan Taichung, Taiwan Tainan, Taiwan (2)
	Visit **portal.guardanthealth.com** for trials not within the same state as the physician's office			
TP53 R267W	NCT04768868 Jian Wang,Jian. wang@impacttherapeutics.com,+86 18613056501	The Safety and Pharmacokinetics Preliminary Efficacy of IMP7068 in Patients With Advanced Solid Tumors	Phase 1	Taipei, Taiwan Taoyuan, Taiwan Taichung, Taiwan Tainan, Taiwan (2)
	Visit **portal.guardanthealth.com** for trials not within the same state as the physician's office			
TP53 R249fs	NCT04768868 Jian Wang,Jian. wang@impacttherapeutics.com,+86 18613056501	The Safety and Pharmacokinetics Preliminary Efficacy of IMP7068 in Patients With Advanced Solid Tumors	Phase 1	Taipei, Taiwan Taoyuan, Taiwan Taichung, Taiwan Tainan, Taiwan (2)
	Visit **portal.guardanthealth.com** for trials not within the same state as the physician's office			
TP53 L265del	NCT04768868 Jian Wang,Jian. wang@impacttherapeutics.com,+86 18613056501	The Safety and Pharmacokinetics Preliminary Efficacy of IMP7068 in Patients With Advanced Solid Tumors	Phase 1	Taipei, Taiwan Taoyuan, Taiwan Taichung, Taiwan Tainan, Taiwan (2)
	Visit **portal.guardanthealth.com** for trials not within the same state as the physician's office			
KRAS G12D	Visit **portal.guardanthealth.com** for trials not within the same state as the physician's office			

More clinical trial options available at **portal.guardanthealth.com**

Definitions

Deletion (Del): The following alteration was detected in this patient: *TP53* L265del, R249fs. Guardant360 detects short deletions in exons of certain genes (see Table 1), including potential splice site-disrupting events.

Interpretation

Somatic alterations were detected in the circulating cell-free DNA isolated from this patient's blood specimen. These genomic alterations are cancer-associated somatic variants, some of which have been associated with either increased or reduced clinical response to specific treatments. The percentage of altered cell-free DNA circulating (% cfDNA) in blood is related to the unique tumor biology of each patient. Factors that may affect the % cfDNA of detected somatic alterations include tumor growth, turn over, size, heterogeneity, vascularization, disease progression, and treatment.

Method and Limitations

Guardant360 sequences 74 cancer-associated genes to identify somatic alterations. Cell-free DNA (cfDNA) is extracted from plasma, enriched for targeted regions, and sequenced using the Illumina platform and hg19 as the reference genome. All exons are sequenced in some genes; only clinically significant exons are sequenced in other genes. The types of genomic alterations detected by Guardant360 include single nucleotide variants, gene amplifications, fusions, short insertions/deletions (longest detected, 70 base pairs), and splice site disrupting events (see Table 1). Microsatellite Instability (MSI) is assessed for all cancer types by evaluating somatic changes in the length of repetitive sequences on the Guardant360 panel. A "Not Detected" result in samples where the highest % cfDNA is < 0.2% is an inconclusive result because it does not preclude MSI-High status in tissue. MSI status is currently not reported for earlier panel versions. This version of the Guardant360 test is not validated for the detection of other types of genomic alterations, such as complex rearrangements or gene deletions. Certain sample or variant characteristics, such as low cfDNA concentration, may result in reduced analytic sensitivity. Guardant360 cannot discern the source of circulating cfDNA, and for some variants in the range of ~40 to 60% cfDNA, the test cannot easily distinguish germline variants from somatic alterations. Guardant360 is not validated for the detection of germline or de novo variants that are associated with hereditary cancer risk. Tissue genotyping should be considered when plasma genotyping is negative, if clinically appropriate.

Table 1: Genes on the Guardant360 Panel

Guardant360 reports single nucleotide variants, splice site mutations, and insertion and deletion variants (indels) in all clinically relevant exons in 74 genes and reports other variant types in select genes as indicated below.

AKT1	*ALK* #	*APC*	*AR* †	*ARAF*	*ARID1A*	*ATM*	*BRAF* †	*BRCA1*
BRCA2	*CCND1* †	*CCND2* †	*CCNE1* †	*CDH1*	*CDK12*	*CDK4* †	*CDK6* †	*CDKN2A*
CTNNB1	*DDR2*	*EGFR* †	*ERBB2* †	*ESR1*	*EZH2*	*FBXW7*	*FGFR1* †	*FGFR2* # †
FGFR3 #	*GATA3*	*GNA11*	*GNAQ*	*GNAS*	*HNF1A*	*HRAS*	*IDH1*	*IDH2*
JAK2	*JAK3*	*KIT* †	*KRAS* †	*MAP2K1*	*MAP2K2*	*MAPK1*	*MAPK3*	*MET* †
MLH1	*MPL*	*MTOR*	*MYC* †	*NF1*	*NFE2L2*	*NOTCH1*	*NPM1*	*NRAS*
NTRK1 #	*NTRK3*	*PDGFRA* †	*PIK3CA* †	*PTEN*	*PTPN11*	*RAF1* †	*RB1*	*RET* #
RHEB	*RHOA*	*RIT1*	*ROS1* #	*SMAD4*	*SMO*	*STK11*	*TERT* ‡	*TP53*
TSC1	*VHL*							

‡ Guardant360 reports alterations in the promoter region of this gene.

Guardant360 reports fusion events involving this gene.

† Guardant360 reports amplifications of this gene.

About the Test

The Guardant360 assay was developed and its performance characteristics were determined by Guardant Health, Inc. This test has not been cleared or approved by the U.S. Food and Drug Administration (FDA). The FDA has determined that such clearance or approval is not necessary. This test may be used for clinical purposes and should not be regarded as investigational or for research only. Guardant Health's clinical reference laboratory is certified under the Clinical Laboratory Improvement Amendments of 1988 (CLIA) as qualified to perform high complexity clinical laboratory testing. The laboratory report should be interpreted in the context of other clinical information and laboratory, pathology, and imaging studies by a qualified medical professional prior to initiating or changing a patient's treatment plan. The selection of any, all, or none of the drugs associated with potential clinical benefit (or potential lack of clinical benefit) is entirely at the discretion of the treating medical professional. Drug and trial information are based on the diagnosis written on the submitted test request form; this information is not based on any supplemental information provided by the requesting medical professional, including pathology reports or other molecular studies. Some drugs listed in this report may not be approved or cleared by the FDA for the indicated use. Guardant Health makes no endorsement, express or implied, of any product, physician, or procedure contained in this report. This report makes no promises or guarantees that a particular medication will affect (or not affect) the clinical outcome of any patient.

Testing Performed at: Guardant Health

Laboratory Director: Martina Lefterova, MD PhD | CLIA ID: 05D2070300 | CAP #: 8765297 | 505 Penobscot Drive, Redwood City, CA, 94063, USA

Additional information is available

Any therapeutic annotations are based on publicly available information. This information is described in the "Detailed Therapy Results" and "Relevance of Detected Alterations" sections.

Visit **portal.guardanthealth.com** or email **clientservices@guardanthealth.com** with A0779678 in the subject line of the email for:

- Additional clinical trials
- Detailed Therapy Results
- Relevance of Detected Alterations
- References

If you would like to receive this additional information with every Guardant360 report, please call client services at **855.698.8887** to opt-in.

Additional information begins on the next page.

List of Available Clinical Trials

Alteration	Trial ID / Contact	Title	Phase	Site (number in parenthesis is count of trial sites)
TP53 Y220C	NCT03968653 Debiopharm International S.A, clinicaltrials@debiopharm.com,+41 21 321 01 11	Study of Oral Debio 0123 in Combination With Carboplatin in Participants With Advanced Solid Tumors	Phase 1	Spain; Netherlands (3)
	NCT04555837 Faye M Johnson, MD, fmjohns@mdanderson.org,(713) 792-6363	Alisertib and Pembrolizumab for the Treatment of Patients With Rb-deficient Head and Neck Squamous Cell Cancer	Phase 1 /Phase 2	Houston, TX
	NCT04585750 PMV Pharma Clinical Study Information Center,clinicaltrials@pmvpharma.com, (609) 235-4038	The Evaluation of PC14586 in Patients With Advanced Solid Tumors Harboring a p53 Y220C Mutation (PYNNACLE)	Phase 1 /Phase 2	Los Angeles, CA; Seattle, WA; Houston, TX; Cleveland, OH; New York, NY; Austin, TX; New Haven, CT; Portland, OR; San Antonio, TX; Nashville, TN; Boston, MA (2)
	NCT04768868 Jian Wang,Jian. wang@impacttherapeutics.com,+86 18613056501	The Safety and Pharmacokinetics Preliminary Efficacy of IMP7068 in Patients With Advanced Solid Tumors	Phase 1	Louisville, KY; Boston, MA; Atlanta, GA; Dallas, TX; Fairway, KS; San Antonio, TX; China (4); Taiwan (5)
	NCT04869475 Min Shi, MD & Ph. D,sm11998@rjh.com. cn,+86-21-64370045	Arsenic Trioxide in Refractory Solid Tumors With Rescuable p53 Mutation	Phase 2	China
	NCT05109975 Debiopharm International S.A, clinicaltrials@debiopharm.com,+41 21 321 01 11	A Study to Evaluate Safety and Preliminary Anti-tumor Activity of Debio 0123 as Monotherapy in Adult Participants With Advanced Solid Tumors	Phase 1	Grand Rapids, MI; San Antonio, TX; Switzerland (2)
TP53 Y234C	NCT03968653 Debiopharm International S.A, clinicaltrials@debiopharm.com,+41 21 321 01 11	Study of Oral Debio 0123 in Combination With Carboplatin in Participants With Advanced Solid Tumors	Phase 1	Spain; Netherlands (3)
	NCT04555837 Faye M Johnson, MD, fmjohns@mdanderson.org,(713) 792-6363	Alisertib and Pembrolizumab for the Treatment of Patients With Rb-deficient Head and Neck Squamous Cell Cancer	Phase 1 /Phase 2	Houston, TX
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	NCT05109975 Debiopharm International S.A, clinicaltrials@debiopharm.com,+41 21 321 01 11	A Study to Evaluate Safety and Preliminary Anti-tumor Activity of Debio 0123 as Monotherapy in Adult Participants With Advanced Solid Tumors	Phase 1	Grand Rapids, MI; San Antonio, TX; Switzerland (2)
TP53 R267W	NCT03968653 Debiopharm International S.A, clinicaltrials@debiopharm.com,+41 21 321 01 11	Study of Oral Debio 0123 in Combination With Carboplatin in Participants With Advanced Solid Tumors	Phase 1	Spain; Netherlands (3)
	NCT04555837 Faye M Johnson, MD, fmjohns@mdanderson.org,(713) 792-6363	Alisertib and Pembrolizumab for the Treatment of Patients With Rb-deficient Head and Neck Squamous Cell Cancer	Phase 1 /Phase 2	Houston, TX
	NCT04768868 Jian Wang,Jian. wang@impacttherapeutics.com,+86 18613056501	The Safety and Pharmacokinetics Preliminary Efficacy of IMP7068 in Patients With Advanced Solid Tumors	Phase 1	Louisville, KY; Boston, MA; Atlanta, GA; Dallas, TX; Fairway, KS; San Antonio, TX; China (4); Taiwan (5)

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	NCT04869475 Min Shi, MD & Ph. D,sm11998@rjh.com.cn,+86-21-64370045	Arsenic Trioxide in Refractory Solid Tumors With Rescuable p53 Mutation	Phase 2	China
	NCT05109975 Debiopharm International S.A, clinicaltrials@debiopharm.com,+41 21 321 01 11	A Study to Evaluate Safety and Preliminary Anti-tumor Activity of Debio 0123 as Monotherapy in Adult Participants With Advanced Solid Tumors	Phase 1	Grand Rapids, MI; San Antonio, TX; Switzerland (2)
TP53 R249fs	NCT03968653 Debiopharm International S.A, clinicaltrials@debiopharm.com,+41 21 321 01 11	Study of Oral Debio 0123 in Combination With Carboplatin in Participants With Advanced Solid Tumors	Phase 1	Spain; Netherlands (3)
	NCT04555837 Faye M Johnson, MD, fmjohns@mdanderson.org,(713) 792-6363	Alisertib and Pembrolizumab for the Treatment of Patients With Rb-deficient Head and Neck Squamous Cell Cancer	Phase 1 /Phase 2	Houston, TX
	NCT04768868 Jian Wang,Jian.wang@impacttherapeutics.com,+86 18613056501	The Safety and Pharmacokinetics Preliminary Efficacy of IMP7068 in Patients With Advanced Solid Tumors	Phase 1	Louisville, KY; Boston, MA; Atlanta, GA; Dallas, TX; Fairway, KS; San Antonio, TX; China (4); Taiwan (5)
	NCT04869475 Min Shi, MD & Ph. D,sm11998@rjh.com.cn,+86-21-64370045	Arsenic Trioxide in Refractory Solid Tumors With Rescuable p53 Mutation	Phase 2	China
	NCT05109975 Debiopharm International S.A, clinicaltrials@debiopharm.com,+41 21 321 01 11	A Study to Evaluate Safety and Preliminary Anti-tumor Activity of Debio 0123 as Monotherapy in Adult Participants With Advanced Solid Tumors	Phase 1	Grand Rapids, MI; San Antonio, TX; Switzerland (2)
TP53 L265del	NCT03968653 Debiopharm International S.A, clinicaltrials@debiopharm.com,+41 21 321 01 11	Study of Oral Debio 0123 in Combination With Carboplatin in Participants With Advanced Solid Tumors	Phase 1	Spain; Netherlands (3)
	NCT04555837 Faye M Johnson, MD, fmjohns@mdanderson.org,(713) 792-6363	Alisertib and Pembrolizumab for the Treatment of Patients With Rb-deficient Head and Neck Squamous Cell Cancer	Phase 1 /Phase 2	Houston, TX
	NCT04768868 Jian Wang,Jian.wang@impacttherapeutics.com,+86 18613056501	The Safety and Pharmacokinetics Preliminary Efficacy of IMP7068 in Patients With Advanced Solid Tumors	Phase 1	Louisville, KY; Boston, MA; Atlanta, GA; Dallas, TX; Fairway, KS; San Antonio, TX; China (4); Taiwan (5)
	NCT04869475 Min Shi, MD & Ph. D,sm11998@rjh.com.cn,+86-21-64370045	Arsenic Trioxide in Refractory Solid Tumors With Rescuable p53 Mutation	Phase 2	China
	NCT05109975 Debiopharm International S.A, clinicaltrials@debiopharm.com,+41 21 321 01 11	A Study to Evaluate Safety and Preliminary Anti-tumor Activity of Debio 0123 as Monotherapy in Adult Participants With Advanced Solid Tumors	Phase 1	Grand Rapids, MI; San Antonio, TX; Switzerland (2)
KRAS G12D	NCT03745326 NCI SB Immunotherapy Recruitment Center,irc@nih.gov,(866) 820-4505	Administering Peripheral Blood Lymphocytes Transduced With a Murine T-Cell Receptor Recognizing the G12D Variant of Mutated RAS in HLA-A*11:01 Patients	Phase 1 /Phase 2	Bethesda, MD
	NCT05382559 Astellas Pharma Inc.,astellas.registration@astellas.com,800-888-7704	A Study of ASP3082 in Adults With Previously Treated Solid Tumors	Phase 1	Westwood, KS; Saint Louis, MO; Boston, MA; New Haven, CT; San Antonio, TX; Sarasota, FL; Santa Monica, CA; Fairfax, VA; New York, NY (2); Nashville, TN (2)
	NCT05554328 See https://clinicaltrials.gov/show	Testing the Use of the Combination of Selumetinib and Olaparib or Selumetinib Alone	Phase 2	Kingman, AZ; Harrison, NY; Bethesda, MD; Wynnewood, PA;

List of Available Clinical Trials

Alteration	Trial ID / Contact	Title	Phase	Site (number in parenthesis is count of trial sites)
	/NCT05554328	Targeted Treatment for RAS Pathway Mutant Recurrent or Persistent Ovarian and Endometrial Cancers, A ComboMATCH Treatment Trial		Commack, NY; Arroyo Grande, CA; New York, NY; Middletown, NJ; Paoli, PA; Renton, WA; Bryn Mawr, PA; Ontario, OR; Media, PA; Cumberland, MD; Las Vegas, NV (2); MT (6); OH (16); ID (11); MI (14)
	NCT05726864 Clinical Trial Inquiries, clinicaltrialinquiries@elicio.com,617-714-9884	A Study of ELI-002 7P in Subjects With KRAS /NRAS Mutated Solid Tumors	Phase 1 /Phase 2	Los Angeles, CA; Houston, TX; Duarte, CA; New York, NY; Lake Success, NY; Gainesville, FL
	NCT05737706 Mirati Therapeutics Study Locator Services, miratistudylocator@careboxhealth.com, 18448935530	Study of MRTX1133 in Patients With Advanced Solid Tumors Harboring a KRAS G12D Mutation	Phase 1 /Phase 2	Orlando, FL; Houston, TX; Grand Rapids, MI; New York, NY; Baltimore, MD; Nashville, TN; Fairfax, VA; Boston, MA (2); San Antonio, TX (2)

Detailed Therapy Results

Alteration	Drug	Trade Name	Target	Current Status
KRAS G12D	Anti-KRAS G12D mTCR cells		Peripheral blood lymphocytes transduced with a murine T-Cell receptor recognizing K-Ras G12D.	Phase 2 (Solid Tumor)
	ASN007		ERK1/2 kinase inhibitor.	Phase 1 (Solid Tumor)
	ASP3082		KRAS G12D degrader.	Phase 1 (Solid Tumor) Phase 1 (Brain and Central Nervous System Tumors)
	ASTX029		ERK1/2 kinase inhibitor.	Phase 2 (Solid Tumor)
	BBP-398		Shp-2 inhibitor.	Phase 1 (Solid Tumor)
	BI 1701963		Pan-K-Ras inhibitor targeting the interaction of K-Ras and SOS-1.	Phase 1 (Solid Tumor) Phase 1 (Brain and Central Nervous System Tumors)
	BI 3011441		MEK1,2 inhibitor.	Phase 1 (Solid Tumor)
	Binimetinib	Mektovi	MEK1,2 inhibitor.	Phase 1 (Solid Tumor) FDA Approved in other indications (Melanoma with BRAF V600E/K mutation)
	BMF-219		Covalent menin inhibitor.	Phase 2 (Diabetes)
	Cobimetinib	Cotellic	MEK1,2 inhibitor.	Phase 2 (Solid Tumor) FDA Approved in other indications (Melanoma with BRAF V600E/K mutation, Histiocytic and dendritic cell neoplasms)
	CS3006		MEK1,2 inhibitor.	Phase 1 (Solid Tumor)
	Defactinib		Focal adhesion kinase (FAK) inhibitor, upstream of MAPK pathway.	Phase 2 (Solid Tumor) Phase 2 (Lymphoma, Multiple myeloma (MM))
	E6201		MEK1,2, Mekk1, and Flt3 inhibitor.	Phase 1 (Solid Tumor) Phase 1 (Acute myeloid leukemia (AML))
	ELI-002		Amph-modified KRAS G12D and G12R peptide immunotherapy.	Phase 1 (Solid Tumor)
	ERAS-601		Shp-2 inhibitor.	Phase 1 (Solid Tumor)
	GI-4000		Mutant K-Ras vaccine.	Phase 2 (Pancreatic carcinoma, Non-small cell lung carcinoma (NSCLC), Colorectal carcinoma (CRC))
	HH2710		ERK1/2 kinase inhibitor.	Phase 2 (Solid Tumor)
	HMPL-295		ERK1/2 kinase inhibitor.	Phase 1 (Solid Tumor)
	JAB-3068		Shp-2 inhibitor.	Phase 1 (Solid Tumor) Phase 2 (Non-small cell lung carcinoma (NSCLC), Head and neck squamous cell carcinoma (HNSCC), Esophageal carcinoma)
	JAB-3312		Shp-2 inhibitor.	Phase 1 (Solid Tumor) Phase 1 (Brain and Central Nervous System Tumors)
	KO-947		ERK1/2 kinase inhibitor.	Phase 1 (Solid Tumor)
	LTT462		ERK1/2 kinase inhibitor.	Phase 1 (Solid Tumor)
	Mesenchymal stromal cells-derived exosomes		Mesenchymal stromal cells-derived exosomes with KRAS G12D siRNA.	Phase 1 (Pancreatic carcinoma)

Detailed Therapy Results

Alteration	Drug	Trade Name	Target	Current Status
	with KRAS G12D siRNA			
	Mirdametinib		MEK1,2 inhibitor.	Phase 1 (Solid Tumor) Phase 2 (Glioma, Non-small cell lung carcinoma (NSCLC), Neurofibroma, Breast carcinoma, Glioneuronal tumor, Neurofibromatosis type 1, Lung cancer, Colorectal carcinoma (CRC))
	MK-8353		ERK1/2 kinase inhibitor.	Phase 1 (Solid Tumor)
	MRTX1133		KRAS G12D inhibitor.	Phase 1 (Solid Tumor)
	PF-07284892		Shp-2 inhibitor.	Phase 1 (Solid Tumor) Phase 1 (Brain and Central Nervous System Tumors)
	PF-07799544		MEK Brain Penetrant Inhibitor.	Phase 1 (Glioma, Melanoma, Non-small cell lung carcinoma (NSCLC), Thyroid carcinoma, Colorectal carcinoma (CRC))
	Pimasertib		MEK1,2 inhibitor.	Phase 1 (Solid Tumor) Phase 2 (Melanoma, Pancreatic ductal adenocarcinoma, Ovarian carcinoma, Hematologic malignancies, Colorectal carcinoma (CRC))
	Refametinib		MEK1,2 inhibitor.	Phase 1 (Solid Tumor) Phase 2 (Gallbladder carcinoma, Hepatocellular carcinoma (HCC), Pancreatic carcinoma, Breast carcinoma, Cholangiocarcinoma, Colorectal carcinoma (CRC), Combined hepatocellular carcinoma and cholangiocarcinoma)
	RMC-4630		Shp-2 inhibitor.	Phase 2 (Solid Tumor)
	RMC-6236		Multispecific K-Ras inhibitor.	Phase 1 (Solid Tumor)
	Selumetinib	Koselugo	MEK1,2 inhibitor.	Phase 2 (Endometrial carcinoma) FDA Approved in other indications (NF1-related plexiform neurofibroma)
	SHR7390		MEK1,2 inhibitor.	Phase 1 (Solid Tumor)
	siG12D-LODER		Antisense oligonucleotide targeting G12D mutant K-Ras.	Phase 2 (Pancreatic carcinoma)
	Temuterkib		ERK1/2 kinase inhibitor.	Phase 1 (Solid Tumor) Phase 1 (Pancreatic ductal adenocarcinoma)
	Tizaterkib		ERK1/2 kinase inhibitor.	Phase 1 (Solid Tumor) Phase 1 (Hematologic malignancies)
	Trametinib	Mekinist	MEK1,2 inhibitor.	Phase 1 (Endometrial carcinoma) FDA Approved in other indications (Melanoma with BRAF V600 mutation)
	Ulixertinib		ERK1/2 kinase inhibitor.	Phase 1 (Solid Tumor) Phase 2 (Gastric carcinoma, Melanoma, Histiocytic and dendritic cell neoplasms, Langerhans cell histiocytosis (LCH), Uveal melanoma, Acute myeloid leukemia (AML), Gastrointestinal carcinoma, Non-Hodgkin lymphoma (NHL), Myelodysplastic Syndrome (MDS))
	V941		Mutant K-Ras vaccine.	Phase 1 (Pancreatic carcinoma, Non-small cell lung carcinoma (NSCLC), Colorectal carcinoma (CRC))

Detailed Therapy Results

Alteration	Drug	Trade Name	Target	Current Status
TP53 Y220C Y234C R249fs L265del R267W	Adavosertib		Wee1 tyrosine kinase inhibitor.	Phase 2 (Endometrial carcinoma) Phase 2 (Embryonal tumor with multi-layered rosettes (ETMR), Medulloblastoma, Small cell lung carcinoma (SCLC), Solid Tumor, Primary myelofibrosis (PMF), Ovarian carcinosarcoma, Acute myeloid leukemia (AML), MDS/MPN, unclassifiable, Chronic myelomonocytic leukemia (CMML), Peritoneal papillary serous carcinoma, Myelodysplastic Syndrome (MDS))
	Alisertib		Aurora A kinase inhibitor.	Phase 1 (Solid Tumor) Phase 3 (Peripheral T-cell lymphoma (PTCL))
	AMG 900		Aurora A/B/C kinase inhibitor.	Phase 1 (Solid Tumor) Phase 1 (Acute myeloid leukemia (AML))
	AT9283		Aurora A, B, Jak2, Jak3, Bcr-Abl kinase inhibitor.	Phase 1 (Solid Tumor) Phase 2 (Acute myeloid leukemia (AML), Multiple myeloma (MM), Acute lymphoblastic leukemia (ALL))
	ATO	Trisenox	PML-RARA inhibitor. Inhibits multiple signaling pathways, including the Hedgehog pathway.	FDA Approved in other indications (Acute myeloid leukemia (AML), Acute promyelocytic leukemia (APL))
	AZD2811		Nanoparticle formulation of Aurora kinase B inhibitor barasertib (AZD1152).	Phase 1 (Solid Tumor) Phase 2 (Acute myeloid leukemia (AML), Myelodysplastic Syndrome (MDS))
	Debio 0123		Wee1 tyrosine kinase inhibitor.	Phase 1 (Solid Tumor)
	ENMD-2076		Aurora A kinase inhibitor.	Phase 1 (Solid Tumor) Phase 2 (Fibrolamellar hepatocellular carcinoma, Ovarian carcinoma, Breast carcinoma, Fallopian tube adenocarcinoma, Sarcoma)
	IMP7068		Wee1 tyrosine kinase inhibitor.	Phase 1 (Solid Tumor)
	Kevetrin		Blocks Mdm2-p53 interaction, restoring transcriptional activity of p53.	Phase 1 (Solid Tumor) Phase 2 (Ovarian carcinoma)
	LY3295668		Aurora A kinase inhibitor.	Phase 1 (Solid Tumor) Phase 2 (Head and neck squamous cell carcinoma (HNSCC), Small cell lung carcinoma (SCLC), Breast carcinoma (triple negative), Breast carcinoma (hormone receptor +, HER2-))
	SGT-53		TP53 gene therapy delivered via transferrin-targeted nanoparticles.	Phase 1 (Solid Tumor) Phase 2 (Glioblastoma, Glioma, Pancreatic carcinoma)
	SNS-314		Aurora A/B kinase inhibitor.	Phase 1 (Solid Tumor)
	SY-4835		Wee1 tyrosine kinase inhibitor.	Phase 1 (Solid Tumor)
	TAS-119		Aurora A kinase inhibitor.	Phase 1 (Solid Tumor)
	TT-00420		Aurora A/B kinase inhibitor.	Phase 1 (Solid Tumor) Phase 1 (Breast carcinoma (triple negative))
TP53 Y220C Y234C L265del R267W	ALT-801		p53-targeted T-cell receptor-IL2 fusion.	Phase 1 (Solid Tumor) Phase 2 (Melanoma, Urothelial carcinoma, Bladder carcinoma, Urethral melanoma, Urethral carcinoma, Multiple myeloma (MM))
	Eprenetapopt		Reactivates mutant p53.	Phase 3 (Myelodysplastic Syndrome (MDS))
TP53	COTI-2		Reactivates mutant p53.	Phase 1 (Endometrial carcinoma) Phase 1 (Head

Detailed Therapy Results

Alteration	Drug	Trade Name	Target	Current Status
Y220C Y234C R267W				and neck squamous cell carcinoma (HNSCC), Ovarian carcinoma, Cervical carcinoma)
TP53 Y220C	PC14586		Reactivator of p53 specific for Y220C mutant protein.	Phase 1 (Solid Tumor)

Relevance of Detected Alterations

Alteration	Role in Disease	Effect on Drug Sensitivity	Effect on Drug Resistance
KRAS G12D	The KRAS gene is one of the most commonly mutated genes in human malignancies, with high incidences in pancreatic, colorectal, and lung cancers. [1-3]. Activating KRAS mutations have been shown to be early events in endometrial carcinogenesis, detected in both endometrial hyperplasia and carcinoma cases. [4-8].	Many of the current attempts to target K-Ras are directed against its downstream signaling pathways, Raf /MEK/ERK and PI3K/Akt/mTOR. [9,10]. Clinical studies have suggested limited efficacy of MEK inhibitors in KRAS mutant tumors; however, combinations of MEK inhibitors with other targeted therapies may still be relevant. [11-20]. Other clinical approaches are being investigated preclinically and clinically in the context of KRAS-mutant tumors, including FAK and Shp-2 inhibitors. [21-26]. In addition, inhibitors specifically targeting KRAS G12C and cell-based therapies targeting KRAS G12V and G12D are being investigated clinically and preclinically. [27-30]. Sotorasib and adagrasib have been FDA-approved in patients with locally advanced or metastatic non-small cell lung carcinoma harboring a KRAS G12C mutation, as determined by an FDA-approved test, following treatment with at least one prior systemic therapy. [31-35]. In addition, the combination of adagrasib with cetuximab has been reported to provide clinical benefit in colorectal carcinoma patients with KRAS G12C mutation and has been granted “breakthrough” designation by the FDA for accelerated review. [36,37].	In some cancer types, such as colorectal cancer (CRC) and non-small cell lung cancer (NSCLC), activating KRAS mutations and KRAS amplification have been associated with resistance to Egfr-targeted therapies. [38-46].
TP53 Y220C	Loss of tumor suppressor p53, which is encoded by the TP53 gene, is common in aggressive advanced cancers. [47]. Carriers of a germline mutation in TP53 have Li-Fraumeni Syndrome, an inherited cancer syndrome resulting in multiple tumors in early adulthood, including breast cancer, brain tumors, and leukemias. [48-50]. Expression of p53 in normal cells is low; however, TP53 alterations, including those that result in loss of p53 tumor suppressor function, may lead to stabilization and increased expression of p53, particularly in the nucleus, and several studies have shown that it may have oncogenic gain-of-function effects. [51-55]. Expression of p53 protein has been correlated with high tumor grade, metastasis, and non-endometrioid histology in endometrial carcinoma cases. [56-61].	At present, there are no approved therapies targeting TP53 alterations, despite their high prevalence in cancer. Therapeutic approaches under investigation include gene therapy for TP53 and (dendritic cell-based) TP53 vaccines. [62-64]. Inhibition of components of the DNA damage checkpoint, including Wee1, has been reported to enhance the activity of DNA-damaging agents in preclinical cancer models with deficiency of p53 function. [65-67]. Clinical trials of the Wee1 inhibitor adavosertib (MK-1775) are currently underway for patients with solid tumors and hematologic malignancies. Studies have reported Aurora kinase A to be activated in cells harboring TP53 mutation, and Aurora kinase A and B inhibitors have been reported to activate wild-type p53 in cellular assays; thus, tumors retaining a wild-type TP53 allele may benefit from Aurora kinase inhibitors. [68-73].	Mutations in TP53 may increase resistance to ionizing radiation therapy. [74,75].

Relevance of Detected Alterations

Alteration	Role in Disease	Effect on Drug Sensitivity	Effect on Drug Resistance
TP53 Y234C	Loss of tumor suppressor p53, which is encoded by the TP53 gene, is common in aggressive advanced cancers. [47]. Carriers of a germline mutation in TP53 have Li-Fraumeni Syndrome, an inherited cancer syndrome resulting in multiple tumors in early adulthood, including breast cancer, brain tumors, and leukemias. [48-50]. Expression of p53 in normal cells is low; however, TP53 alterations, including those that result in loss of p53 tumor suppressor function, may lead to stabilization and increased expression of p53, particularly in the nucleus, and several studies have shown that it may have oncogenic gain-of-function effects. [51-55]. Expression of p53 protein has been correlated with high tumor grade, metastasis, and non-endometrioid histology in endometrial carcinoma cases. [56-61].	At present, there are no approved therapies targeting TP53 alterations, despite their high prevalence in cancer. Therapeutic approaches under investigation include gene therapy for TP53 and (dendritic cell-based) TP53 vaccines. [62-64]. Inhibition of components of the DNA damage checkpoint, including Wee1, has been reported to enhance the activity of DNA-damaging agents in preclinical cancer models with deficiency of p53 function. [65-67]. Clinical trials of the Wee1 inhibitor adavosertib (MK-1775) are currently underway for patients with solid tumors and hematologic malignancies. Studies have reported Aurora kinase A to be activated in cells harboring TP53 mutation, and Aurora kinase A and B inhibitors have been reported to activate wild-type p53 in cellular assays; thus, tumors retaining a wild-type TP53 allele may benefit from Aurora kinase inhibitors. [68-73].	Mutations in TP53 may increase resistance to ionizing radiation therapy. [74,75].
TP53 R267W	Loss of tumor suppressor p53, which is encoded by the TP53 gene, is common in aggressive advanced cancers. [47]. Carriers of a germline mutation in TP53 have Li-Fraumeni Syndrome, an inherited cancer syndrome resulting in multiple tumors in early adulthood, including breast cancer, brain tumors, and leukemias. [48-50]. Expression of p53 in normal cells is low; however, TP53 alterations, including those that result in loss of p53 tumor suppressor function, may lead to stabilization and increased expression of p53, particularly in the nucleus, and several studies have shown that it may have oncogenic gain-of-function effects. [51-55]. Expression of p53 protein has been correlated with high tumor grade, metastasis, and non-endometrioid histology in endometrial carcinoma cases. [56-61].	At present, there are no approved therapies targeting TP53 alterations, despite their high prevalence in cancer. Therapeutic approaches under investigation include gene therapy for TP53 and (dendritic cell-based) TP53 vaccines. [62-64]. Inhibition of components of the DNA damage checkpoint, including Wee1, has been reported to enhance the activity of DNA-damaging agents in preclinical cancer models with deficiency of p53 function. [65-67]. Clinical trials of the Wee1 inhibitor adavosertib (MK-1775) are currently underway for patients with solid tumors and hematologic malignancies. Studies have reported Aurora kinase A to be activated in cells harboring TP53 mutation, and Aurora kinase A and B inhibitors have been reported to activate wild-type p53 in cellular assays; thus, tumors retaining a wild-type TP53 allele may benefit from Aurora kinase inhibitors. [68-73].	Mutations in TP53 may increase resistance to ionizing radiation therapy. [74,75].
TP53 R249fs	Loss of tumor suppressor p53, which is encoded by the TP53 gene, is common in aggressive advanced cancers. [47]. Carriers of a germline mutation in TP53 have Li-Fraumeni Syndrome, an inherited cancer syndrome resulting in multiple tumors in early adulthood, including breast cancer, brain tumors, and leukemias. [48-50]. Expression of p53 in normal cells is low; however, TP53 alterations,	At present, there are no approved therapies targeting TP53 alterations, despite their high prevalence in cancer. Therapeutic approaches under investigation include gene therapy for TP53 and (dendritic cell-based) TP53 vaccines. [62-64]. Inhibition of components of the DNA damage checkpoint, including Wee1, has been reported to enhance the activity of DNA-damaging agents in preclinical	Mutations in TP53 may increase resistance to ionizing radiation therapy. [74,75].

Relevance of Detected Alterations

Alteration	Role in Disease	Effect on Drug Sensitivity	Effect on Drug Resistance
	including those that result in loss of p53 tumor suppressor function, may lead to stabilization and increased expression of p53, particularly in the nucleus, and several studies have shown that it may have oncogenic gain-of-function effects. [51-55]. Expression of p53 protein has been correlated with high tumor grade, metastasis, and non-endometrioid histology in endometrial carcinoma cases. [56-61].	cancer models with deficiency of p53 function. [65-67]. Clinical trials of the Wee1 inhibitor adavosertib (MK-1775) are currently underway for patients with solid tumors and hematologic malignancies. Studies have reported Aurora kinase A to be activated in cells harboring TP53 mutation, and Aurora kinase A and B inhibitors have been reported to activate wild-type p53 in cellular assays; thus, tumors retaining a wild-type TP53 allele may benefit from Aurora kinase inhibitors. [68-73].	
TP53 L265del	Loss of tumor suppressor p53, which is encoded by the TP53 gene, is common in aggressive advanced cancers. [47]. Carriers of a germline mutation in TP53 have Li-Fraumeni Syndrome, an inherited cancer syndrome resulting in multiple tumors in early adulthood, including breast cancer, brain tumors, and leukemias. [48-50]. Expression of p53 in normal cells is low; however, TP53 alterations, including those that result in loss of p53 tumor suppressor function, may lead to stabilization and increased expression of p53, particularly in the nucleus, and several studies have shown that it may have oncogenic gain-of-function effects. [51-55]. Expression of p53 protein has been correlated with high tumor grade, metastasis, and non-endometrioid histology in endometrial carcinoma cases. [56-61].	At present, there are no approved therapies targeting TP53 alterations, despite their high prevalence in cancer. Therapeutic approaches under investigation include gene therapy for TP53 and (dendritic cell-based) TP53 vaccines. [62-64]. Inhibition of components of the DNA damage checkpoint, including Wee1, has been reported to enhance the activity of DNA-damaging agents in preclinical cancer models with deficiency of p53 function. [65-67]. Clinical trials of the Wee1 inhibitor adavosertib (MK-1775) are currently underway for patients with solid tumors and hematologic malignancies. Studies have reported Aurora kinase A to be activated in cells harboring TP53 mutation, and Aurora kinase A and B inhibitors have been reported to activate wild-type p53 in cellular assays; thus, tumors retaining a wild-type TP53 allele may benefit from Aurora kinase inhibitors. [68-73].	Mutations in TP53 may increase resistance to ionizing radiation therapy. [74,75].

References

1. Farber L, Efrati E, Elkin H, Peerless Y, Sabo E, Ben-Izhak O, Hershkovitz D "Molecular morphometric analysis shows relative intra-tumoural homogeneity for KRAS mutations in colorectal cancer." Virchows Archiv : an international journal of pathology(2011): 487-93
2. Feldmann G, Beaty R, Hruban R, Maitra A "Molecular genetics of pancreatic intraepithelial neoplasia." Journal of hepato-biliary-pancreatic surgery(2007): 224-32
3. Han C, Ma J, Zhao J, Zhou Y, Jing W, Zou H "EGFR mutations, gene amplification, and protein expression and KRAS mutations in primary and metastatic tumors of nonsmall cell lung cancers and their clinical implications: a meta-analysis." Cancer investigation(2011): 626-34
4. Enomoto T, Inoue M, Perantoni A, Buzard G, Miki H, Tanizawa O, Rice J "K-ras activation in premalignant and malignant epithelial lesions of the human uterus." Cancer research(1991): 5308-14
5. Mutter G, Wada H, Faquin W, Enomoto T "K-ras mutations appear in the premalignant phase of both microsatellite stable and unstable endometrial carcinogenesis." Molecular pathology : MP(1999): 257-62
6. Dobrzycka B, Terlikowski S, Mazurek A, Kowalczuk O, Niklińska W, Chyczewski L, Kulikowski M "Mutations of the KRAS oncogene in endometrial hyperplasia and carcinoma." Folia histochemica et cytobiologica(2009): 65-8
7. Zauber P, Denehy T, Taylor R, Ongcapin E, Marotta S, Sabbath-Solitare M "Strong correlation between molecular changes in endometrial carcinomas and concomitant hyperplasia." International journal of gynecological cancer : official journal of the International Gynecological Cancer Society(2015): 863-8
8. Berg A, Hoivik E, Mjøs S, Holst F, Werner H, Tangen I, Taylor-Weiner A, Gibson W, Kusonmano K, Wik E, Trovik J, Halle M, Øyan A, Kalland K, Cherniack A, Beroukhim R, Stefansson I, Mills G, Krakstad C, Salvesen H "Molecular profiling of endometrial carcinoma precursor, primary and metastatic lesions suggests different targets for treatment in obese compared to non-obese patients." Oncotarget(2015): 1327-39
9. Yeh J, Routh E, Rubinas T, Peacock J, Martin T, Shen X, Sandler R, Kim H, Keku T, Der C "KRAS/BRAF mutation status and ERK1/2 activation as biomarkers for MEK1/2 inhibitor therapy in colorectal cancer." Molecular cancer therapeutics(2009): 834-43
10. Britten C "PI3K and MEK inhibitor combinations: examining the evidence in selected tumor types." Cancer chemotherapy and pharmacology(2013): 1395-409
11. Jänne P, van den Heuvel M, Barlesi F, Cobo M, Mazieres J, Crinò L, Orlov S, Blackhall F, Wolf J, Garrido P, Poltoratskiy A, Mariani G, Ghiorghiu D, Kilgour E, Smith P, Kohlmann A, Carlile D, Lawrence D, Bowen K, Vansteenkiste J "Selumetinib Plus Docetaxel Compared With Docetaxel Alone and Progression-Free Survival in Patients With KRAS-Mutant Advanced Non-Small Cell Lung Cancer: The SELECT-1 Randomized Clinical Trial." JAMA(2017): 1844-1853
12. Puyol M, Martín A, Dubus P, Mulero F, Pizcueta P, Khan G, Guerra C, Santamaría D, Barbacid M "A synthetic lethal interaction between K-Ras oncogenes and Cdk4 unveils a therapeutic strategy for non-small cell lung carcinoma." Cancer cell(2010): 63-73
13. Corcoran R, Cheng K, Hata A, Faber A, Ebi H, Coffee E, Greninger P, Brown R, Godfrey J, Cohoon T, Song Y, Lifshits E, Hung K, Shioda T, Dias-Santagata D, Singh A, Settleman J, Benes C, Mino-Kenudson M, Wong K, Engelman J "Synthetic lethal interaction of combined BCL-XL and MEK inhibition promotes tumor regressions in KRAS mutant cancer models." Cancer cell(2013): 121-8
14. Adjei A, Cohen R, Franklin W, Morris C, Wilson D, Molina J, Hanson L, Gore L, Chow L, Leong S, Maloney L, Gordon G, Simmons H, Marlow A, Litwiler K, Brown S, Poch G, Kane K, Haney J, Eckhardt S "Phase I pharmacokinetic and pharmacodynamic study of the oral, small-molecule mitogen-activated protein kinase kinase 1/2 inhibitor AZD6244 (ARRY-142886) in patients with advanced cancers." Journal of clinical oncology : official journal of the American Society of Clinical Oncology(2008): 2139-46
15. Manchado E, Weissmueller S, Morris J, Chen C, Wullenkord R, Lujambio A, de Stanchina E, Poirier J, Gainor J, Corcoran R, Engelman J, Rudin C, Rosen N, Lowe S "A combinatorial strategy for treating KRAS-mutant lung cancer." Nature(2016): 647-51
16. Infante J, Somer B, Park J, Li C, Scheulen M, Kasubhai S, Oh D, Liu Y, Redhu S, Steplewski K, Le N "A randomised, double-blind, placebo-controlled trial of trametinib, an oral MEK inhibitor, in combination with gemcitabine for patients with untreated metastatic adenocarcinoma of the pancreas." European journal of cancer (Oxford, England : 1990)(2014): 2072-81
17. Lito P, Saborowski A, Yue J, Solomon M, Joseph E, Gadal S, Saborowski M, Kastenhuber E, Fellmann C, Ohara K, Morikami K, Miura T, Lukacs C, Ishii N, Lowe S, Rosen N "Disruption of CRAF-mediated MEK activation is required for effective MEK inhibition in KRAS mutant tumors." Cancer cell(2014): 697-710
18. Hochster H, Uboha N, Messersmith W, Gold P, ONeil B, Cohen D, Denlinger C, Cohen S, Leichman C, Leichman L, Lenz H "Phase II study of selumetinib (AZD6244, ARRY-142886) plus irinotecan as second-line therapy in patients with K-RAS mutated colorectal cancer." Cancer chemotherapy and pharmacology(2015): 17-23
19. Blumenschein G, Smit E, Planchard D, Kim D, Cadranel J, De Pas T, Dunphy F, Udud K, Ahn M, Hanna N, Kim J, Mazieres J, Kim S, Baas P, Rappold E, Redhu S, Puski A, Wu F, Jänne P "A randomized phase II study of the MEK1/MEK2 inhibitor trametinib (GSK1120212) compared with docetaxel in KRAS-mutant advanced non-small-cell lung cancer (NSCLC)†." Annals of oncology : official journal of the European Society for Medical Oncology(2015): 894-901
20. Zhu Z, Aref A, Cohoon T, Barbie T, Imamura Y, Yang S, Moody S, Shen R, Schinzel A, Thai T, Reibel J, Tamayo P, Godfrey J, Qian Z, Page A, Maciag K, Chan E, Silkworth W, Labowsky M, Rozhansky L, Mesirov J, Gillanders W, Ogino S, Hacohen N, Gaudet S, Eck M, Engelman J, Corcoran R, Wong K, Hahn W, Barbie D "Inhibition of KRAS-driven tumorigenicity by interruption of an autocrine cytokine circuit." Cancer discovery(2014): 452-65
21. Gerber D, Camidge D, Morgensztern D, Cetnar J, Kelly R, Ramalingam S, Spigel D, Jeong W, Scaglioni P, Zhang S, Li M, Weaver D, Vaikus L, Keegan M, Horobin J, Burns T "Phase 2 study of the focal adhesion kinase inhibitor defactinib (VS-6063) in previously treated advanced KRAS mutant non-small cell lung cancer." Lung cancer (Amsterdam, Netherlands)(2020): 60-67
22. Mainardi S, Mulero-Sánchez A, Prahallad A, Germano G, Bosma A, Krimpenfort P, Lieftink C, Steinberg J, de Wit N, Gonçalves-Ribeiro S, Nadal E, Bardelli A, Villanueva A, Bernards R "SHP2 is required for growth of KRAS-mutant non-small-cell lung cancer in vivo." Nature medicine(2018): 961-967
23. Ruess D, Heynen G, Ciecielski K, Ai J, Berninger A, Kabacaoglu D, Görgülü K, Dantes Z, Wörmann S, Diakopoulos K, Karpathaki A, Kowalska M, Kaya-Aksoy E, Song L, van der Laan E, López-Alberca M, Nazaré M, Reichert M, Saur D, Erkan M, Hopt U, Sainz B, Birchmeier W, Schmid R, Lesina M, Algül H "Mutant KRAS-driven cancers depend on PTPN11/SHP2 phosphatase." Nature medicine(2018): 954-960
24. Konstantinidou G, Ramadori G, Torti F, Kangasniemi K, Ramirez R, Cai Y, Behrens C, Dellinger M, Brekken R, Wistuba I, Heguy A, Teruya-Feldstein J, Scaglioni P "RHOA-FAK is a required signaling axis for the maintenance of KRAS-driven lung adenocarcinomas." Cancer discovery(2013): 444-57
25. Tang K, Constanzo J, Venkateswaran N, Melegari M, Ilcheva M, Morales J, Skoulidis F, Heymach J, Boothman D, Scaglioni P "Focal Adhesion Kinase Regulates the DNA Damage Response and Its Inhibition Radiosensitizes Mutant KRAS Lung Cancer." Clinical cancer research : an official journal of the American Association for Cancer Research(2016): 5851-5863
26. Chen Y, LaMarche M, Chan H, Fekkes P, Garcia-Fortanet J, Acker M, Antonakos B, Chen C, Chen Z, Cooke V, Dobson J, Deng Z, Fei F, Firestone B, Fodor M, Fridrich C, Gao H, Grunenfelder D, Hao H, Jacob J, Ho S, Hsiao K, Kang Z, Karki R, Kato M, Larrow J, La Bonte L, Lenoir F, Liu G, Liu S, Majumdar D, Meyer M, Palermo M, Perez L,

References

Pu M, Price E, Quinn C, Shakya S, Shultz M, Slisz J, Venkatesan K, Wang P, Warmuth M, Williams S, Yang G, Yuan J, Zhang J, Zhu P, Ramsey T, Keen N, Sellers W, Stams T, Fortin P "Allosteric inhibition of SHP2 phosphatase inhibits cancers driven by receptor tyrosine kinases." Nature(2016): 148-52

27. Janes M, Zhang J, Li L, Hansen R, Peters U, Guo X, Chen Y, Babbar A, Firdaus S, Darjania L, Feng J, Chen J, Li S, Li S, Long Y, Thach C, Liu Y, Zarieh A, Ely T, Kucharski J, Kessler L, Wu T, Yu K, Wang Y, Yao Y, Deng X, Zarrinkar P, Brehmer D, Dhanak D, Lorenzi M, Hu-Lowe D, Patricelli M, Ren P, Liu Y "Targeting KRAS Mutant Cancers with a Covalent G12C-Specific Inhibitor." Cell(2018): 578-589.e17
28. Ostrem J, Peters U, Sos M, Wells J, Shokat K "K-Ras(G12C) inhibitors allosterically control GTP affinity and effector interactions." Nature(2013): 548-51
29. Patricelli M, Janes M, Li L, Hansen R, Peters U, Kessler L, Chen Y, Kucharski J, Feng J, Ely T, Chen J, Firdaus S, Babbar A, Ren P, Liu Y "Selective Inhibition of Oncogenic KRAS Output with Small Molecules Targeting the Inactive State." Cancer discovery(2016): 316-29
30. Nagasaka M, Potugari B, Nguyen A, Sukari A, Azmi A, Ou S "KRAS Inhibitors- yes but what next? Direct targeting of KRAS- vaccines, adoptive T cell therapy and beyond." Cancer treatment reviews(2021): 102309
31. "Sotorasib Edges Closer to Approval." Cancer discovery(2021): OF2
32. Li BT, Skoulidis F, Falchook G, et al. "CodeBreaK 100: Registrational Phase 2 Trial of Sotorasib in KRAS p.G12C Mutated Non-small Cell Lung Cancer" Journal of Thoracic Oncology(2020): PS01.07
33. "A Phase 1/2, Open-label Study Evaluating the Safety, Tolerability, Pharmacokinetics, Pharmacodynamics, and Efficacy of AMG 510 Monotherapy in Subjects With Advanced Solid Tumors With KRAS p.G12C Mutation and AMG 510 Combination Therapy in Subjects With Advanced NSCLC With KRAS p.G12C Mutation (CodeBreak 100)" (2021)
34. Riely G, Ou S, Rybkin I, et al. "99O_PR - KRYSTAL-1: Activity and Preliminary Pharmacodynamic (PD) Analysis of Adagrasib (MRTX849) in Patients (Pts) With Advanced Non-Small- Cell Lung Cancer (NSCLC) Harboring KRASG12C Mutation" Annals of Oncology(2021)
35. Jänne P, Riely G, Gadgeel S, Heist R, Ou S, Pacheco J, Johnson M, Sabari J, Leventakos K, Yau E, Bazhenova L, Negrao M, Pennell N, Zhang J, Anderes K, Der-Torossian H, Kheoh T, Velastegui K, Yan X, Christensen J, Chao R, Spira A "Adagrasib in Non-Small-Cell Lung Cancer Harboring a KRASG12C Mutation." The New England journal of medicine(2022): 120-131
36. Yaeger R, Weiss J, Pelster M, Spira A, Barve M, Ou S, Leal T, Bekaii-Saab T, Paweletz C, Heavey G, Christensen J, Velastegui K, Kheoh T, Der-Torossian H, Klempner S "Adagrasib with or without Cetuximab in Colorectal Cancer with Mutated KRAS G12C." The New England journal of medicine(2023): 44-54
37. Klempner S, Weiss J, Pelster M, et al. "KRYSTAL-1: Updated efficacy and safety of adagrasib (MRTX849) with or without cetuximab in patients with advanced colorectal cancer (CRC) harboring a KRASG12C mutation" Annals of Oncology(2022)
38. Favazza L, Parseghian C, Kaya C, Nikiforova M, Roy S, Wald A, Landau M, Proksell S, Dueker J, Johnston E, Brand R, Bahary N, Gorantla V, Rhee J, Pingpank J, Choudry H, Lee K, Paniccia A, Ongchin M, Zureikat A, Bartlett D, Singhi A "KRAS amplification in metastatic colon cancer is associated with a history of inflammatory bowel disease and may confer resistance to anti-EGFR therapy." Modern pathology : an official journal of the United States and Canadian Academy of Pathology, Inc (2020): 1832-1843
39. Eberhard D, Johnson B, Amler L, Goddard A, Heldens S, Herbst R, Ince W, Jänne P, Januario T, Johnson D, Klein P, Miller V, Ostland M, Ramies D, Sebisanovic D, Stinson J, Zhang Y, Seshagiri S, Hillan K "Mutations in the epidermal growth factor receptor and in KRAS are predictive and prognostic indicators in patients with non-small-cell lung cancer treated with chemotherapy alone and in combination with erlotinib." Journal of clinical oncology : official journal of the American Society of Clinical Oncology(2005): 5900-9
40. Linardou H, Dahabreh I, Kanaloupiti D, Siannis F, Bafaloukos D, Kosmidis P, Papadimitriou C, Murray S "Assessment of somatic k-RAS mutations as a mechanism associated with resistance to EGFR-targeted agents: a systematic review and meta-analysis of studies in advanced non-small-cell lung cancer and metastatic colorectal cancer." The Lancet. Oncology(2008): 962-72
41. Ramos F, Macarulla T, Capdevila J, Elez E, Tabernero J "Understanding the predictive role of K-ras for epidermal growth factor receptor-targeted therapies in colorectal cancer." Clinical colorectal cancer(2008): S52-7
42. Campos-Parra A, Zuloaga C, Manríquez M, Avilés A, Borbolla-Escoboza J, Cardona A, Meneses A, Arrieta O "KRAS mutation as the biomarker of response to chemotherapy and EGFR-TKIs in patients with advanced non-small cell lung cancer: clues for its potential use in second-line therapy decision making." American journal of clinical oncology(2015): 33-40
43. De Roock W, De Vriendt V, Normanno N, Ciardiello F, Tejpar S "KRAS, BRAF, PIK3CA, and PTEN mutations: implications for targeted therapies in metastatic colorectal cancer." The Lancet. Oncology(2011): 594-603
44. Douillard J, Oliner K, Siena S, Tabernero J, Burkes R, Barugel M, Humblet Y, Bodoky G, Cunningham D, Jassem J, Rivera F, Kocákova I, Ruff P, Błasińska-Morawiec M, Šmakal M, Canon J, Rother M, Williams R, Rong A, Wiezorek J, Sidhu R, Patterson S "Panitumumab-FOLFOX4 treatment and RAS mutations in colorectal cancer." The New England journal of medicine(2013): 1023-34
45. Valtorta E, Misale S, Sartore-Bianchi A, Nagtegaal I, Paraf F, Lauricella C, Dimartino V, Hobor S, Jacobs B, Ercolani C, Lamba S, Scala E, Veronese S, Laurent-Puig P, Siena S, Tejpar S, Mottolese M, Punt C, Gambacorta M, Bardelli A, Di Nicolantonio F "KRAS gene amplification in colorectal cancer and impact on response to EGFR-targeted therapy." International journal of cancer(2013): 1259-65
46. Li W, Shi Q, Wang W, Liu J, Ren J, Li Q, Hou F "KRAS status and resistance to epidermal growth factor receptor tyrosine-kinase inhibitor treatment in patients with metastatic colorectal cancer: a meta-analysis." Colorectal disease : the official journal of the Association of Coloproctology of Great Britain and Ireland(2014): O370-8
47. Brown C, Lain S, Verma C, Fersht A, Lane D "Awakening guardian angels: drugging the p53 pathway." Nature reviews. Cancer(2009): 862-73
48. Malkin D, Li F, Strong L, Fraumeni J, Nelson C, Kim D, Kassel J, Gryka M, Bischoff F, Tainsky M "Germ line p53 mutations in a familial syndrome of breast cancer, sarcomas, and other neoplasms." Science (New York, N.Y.)(1990): 1233-8
49. Srivastava S, Zou Z, Pirollo K, Blattner W, Chang E "Germ-line transmission of a mutated p53 gene in a cancer-prone family with Li-Fraumeni syndrome." Nature(1991): 747-9
50. Santibáñez-Koref M, Birch J, Hartley A, Jones P, Craft A, Eden T, Crowther D, Kelsey A, Harris M "p53 germline mutations in Li-Fraumeni syndrome." Lancet (London, England)(1991): 1490-1
51. Wang Y, Lin R, Tan Y, Chen J, Chen C, Wang Y "Wild-type p53 overexpression and its correlation with MDM2 and p14ARF alterations: an alternative pathway to non-small-cell lung cancer." Journal of clinical oncology : official journal of the American Society of Clinical Oncology(2005): 154-64
52. Koga T, Hashimoto S, Sugio K, Yoshino I, Nakagawa K, Yonemitsu Y, Sugimachi K, Sueishi K "Heterogeneous distribution of P53 immunoreactivity in human lung adenocarcinoma correlates with MDM2 protein expression, rather than with P53 gene mutation." International journal of cancer(2001): 232-9

References

53. Kato S, Han S, Liu W, Otsuka K, Shibata H, Kanamaru R, Ishioka C "Understanding the function-structure and function-mutation relationships of p53 tumor suppressor protein by high-resolution missense mutation analysis." Proceedings of the National Academy of Sciences of the United States of America(2003): 8424-9
54. Houben R, Hesbacher S, Schmid C, Kauczok C, Flohr U, Haferkamp S, Müller C, Schrama D, Wischhusen J, Becker J "High-level expression of wild-type p53 in melanoma cells is frequently associated with inactivity in p53 reporter gene assays." PloS one(2011): e22096
55. Olivier M, Petitjean A, Marcel V, Pétré A, Mounawar M, Plymoth A, de Fromentel C, Hainaut P "Recent advances in p53 research: an interdisciplinary perspective." Cancer gene therapy(2009): 1-12
56. Jongen V, Briët J, de Jong R, Joppe E, ten Hoor K, Boezen H, Evans D, Hollema H, van der Zee A, Nijman H "Aromatase, cyclooxygenase 2, HER-2/neu, and p53 as prognostic factors in endometrioid endometrial cancer." International journal of gynecological cancer : official journal of the International Gynecological Cancer Society (2009): 670-6
57. Stavropoulos A, Varras M, Vasilakaki T, Varra V, Tsavari A, Varra F, Nonni A, Kavantzas N, Lazaris A "Expression of p53 and PTEN in human primary endometrial carcinomas: Clinicopathological and immunohistochemical analysis and study of their concomitant expression." Oncology letters(2019): 4575-4589
58. Heckl M, Schmoeckel E, Hertlein L, Rottmann M, Jeschke U, Mayr D "The ARID1A, p53 and ß-Catenin statuses are strong prognosticators in clear cell and endometrioid carcinoma of the ovary and the endometrium." PloS one(2018): e0192881
59. Trovik J, Wik E, Werner H, Krakstad C, Helland H, Vandenput I, Njolstad T, Stefansson I, Marcickiewicz J, Tingulstad S, Staff A, Amant F, Akslen L, Salvesen H "Hormone receptor loss in endometrial carcinoma curettage predicts lymph node metastasis and poor outcome in prospective multicentre trial." European journal of cancer (Oxford, England : 1990)(2013): 3431-41
60. Daniilidou K, Frangou-Plemenou M, Grammatikakis J, Grigoriou O, Vitoratos N, Kondi-Pafiti A "Prognostic significance and diagnostic value of PTEN and p53 expression in endometrial carcinoma. A retrospective clinicopathological and immunohistochemical study." Journal of B.U.ON. : official journal of the Balkan Union of Oncology (2013): 195-201
61. Kohler M, Berchuck A, Davidoff A, Humphrey P, Dodge R, Iglehart J, Soper J, Clarke-Pearson D, Bast R, Marks J "Overexpression and mutation of p53 in endometrial carcinoma." Cancer research(1992): 1622-7
62. Schuler P, Harasymczuk M, Visus C, Deleo A, Trivedi S, Lei Y, Argiris A, Gooding W, Butterfield L, Whiteside T, Ferris R "Phase I dendritic cell p53 peptide vaccine for head and neck cancer." Clinical cancer research : an official journal of the American Association for Cancer Research(2014): 2433-44
63. Vermeij R, Leffers N, van der Burg S, Melief C, Daemen T, Nijman H "Immunological and clinical effects of vaccines targeting p53-overexpressing malignancies." Journal of biomedicine & biotechnology(2011): 702146
64. Saito H, Ando S, Morishita N, Lee K, Dator D, Dy D, Shigemura K, Adhim Z, Nibu K, Fujisawa M, Shirakawa T "A combined lymphokine-activated killer (LAK) cell immunotherapy and adenovirus-p53 gene therapy for head and neck squamous cell carcinoma." Anticancer research(2014): 3365-70
65. Ma C, Janetka J, Piwnica-Worms H "Death by releasing the breaks: CHK1 inhibitors as cancer therapeutics." Trends in molecular medicine(2011): 88-96
66. Hirai H, Arai T, Okada M, Nishibata T, Kobayashi M, Sakai N, Imagaki K, Ohtani J, Sakai T, Yoshizumi T, Mizuarai S, Iwasawa Y, Kotani H "MK-1775, a small molecule Wee1 inhibitor, enhances anti-tumor efficacy of various DNA-damaging agents, including 5-fluorouracil." Cancer biology & therapy(2010): 514-22
67. Bridges K, Hirai H, Buser C, Brooks C, Liu H, Buchholz T, Molkentine J, Mason K, Meyn R "MK-1775, a novel Wee1 kinase inhibitor, radiosensitizes p53-defective human tumor cells." Clinical cancer research : an official journal of the American Association for Cancer Research(2011): 5638-48
68. Vilgelm A, Pawlikowski J, Liu Y, Hawkins O, Davis T, Smith J, Weller K, Horton L, McClain C, Ayers G, Turner D, Essaka D, Stewart C, Sosman J, Kelley M, Ecsedy J, Johnston J, Richmond A "Mdm2 and aurora kinase a inhibitors synergize to block melanoma growth by driving apoptosis and immune clearance of tumor cells." Cancer research(2015): 181-93
69. Li Z, Sun Y, Chen X, Squires J, Nowroozizadeh B, Liang C, Huang J "p53 Mutation Directs AURKA Overexpression via miR-25 and FBXW7 in Prostatic Small Cell Neuroendocrine Carcinoma." Molecular cancer research : MCR(2015): 584-91
70. Katayama H, Sen S "Functional significance of Aurora kinase A regulatory interactions with p53-ERα complex in human breast cancer cells." Hormones & cancer(2011): 117-24
71. Tentler J, Ionkina A, Tan A, Newton T, Pitts T, Glogowska M, Kabos P, Sartorius C, Sullivan K, Espinosa J, Eckhardt S, Diamond J "p53 Family Members Regulate Phenotypic Response to Aurora Kinase A Inhibition in Triple-Negative Breast Cancer." Molecular cancer therapeutics(2015): 1117-29
72. Gully C, Velazquez-Torres G, Shin J, Fuentes-Mattei E, Wang E, Carlock C, Chen J, Rothenberg D, Adams H, Choi H, Guma S, Phan L, Chou P, Su C, Zhang F, Chen J, Yang T, Yeung S, Lee M "Aurora B kinase phosphorylates and instigates degradation of p53." Proceedings of the National Academy of Sciences of the United States of America(2012): E1513-22
73. Marxer M, Ma H, Man W, Poon R "p53 deficiency enhances mitotic arrest and slippage induced by pharmacological inhibition of Aurora kinases." Oncogene(2014): 3550-60
74. El-Deiry W "The role of p53 in chemosensitivity and radiosensitivity." Oncogene(2003): 7486-95
75. Miyasaka A, Oda K, Ikeda Y, Sone K, Fukuda T, Inaba K, Makii C, Enomoto A, Hosoya N, Tanikawa M, Uehara Y, Arimoto T, Kuramoto H, Wada-Hiraike O, Miyagawa K, Yano T, Kawana K, Osuga Y, Fujii T "PI3K/mTOR pathway inhibition overcomes radioresistance via suppression of the HIF1-α/VEGF pathway in endometrial cancer." Gynecologic oncology(2015): 174-80